From Client Conversations to Flowcharts

Imagine you have been hired as a software developer. A business owner describes how their business works, what problems they face, and what they want software to do. They will not give you a neatly organized list of requirements.

Your job is to turn that conversation into a clear plan for how the software should work - before writing any code.

Your group will:

- Identify business requirements from a client conversation.
- Determine the information the software needs and what it produces.
- Identify decisions, calculations, actions, and possible outcomes.
- Create a flowchart showing the business logic.
- Test the flowchart using realistic examples.

Part 1 | Understand the client

Read your group's client conversation carefully. Important rules may appear in the middle of a story, out of order, or more than once. Discuss the problem the client wants to solve, what the software must do, and which details are unclear.

Part 2 | Extract the requirements

Write a clear list of the business rules in your own words. Identify inputs (information needed), outputs (results or actions), and questions for the client. If a rule is unclear, do not silently invent an answer: write down the question.

Part 3 | Design the logic

Draw a flowchart showing where the process starts, what decisions are made, how each decision changes the path, what actions or calculations happen, and where the process ends. Label each decision path clearly, such as Yes and No.

Build, Test, and Explain Your Design

Flowchart symbol reference

Oval	Start or end of a process Example: Start Check-In
Rectangle	Action or process Example: Mark Room Occupied
Diamond	Decision with branching paths Example: Is the Room Clean?
Parallelogram	Input or output Example: Enter Reservation Number
Arrow	Direction of flow Example: Continue to next step

Part 4 | Test the flowchart

Walk through your flowchart using at least three situations. For each one, follow the arrows and record what happens.

- Normal case: typical customer or transaction.
- Different outcome: inputs that take another path.
- Edge case: a boundary value, missing information, or overlapping rules.

Part 5 | Share your design

Be ready to explain what the client wanted, which rules your group identified, how the flowchart handles decisions, and at least one question you would ask the client. This is a practice activity, not a graded assignment.

Group checklist

- [ ] Read and discussed the client conversation
- [ ] Listed requirements, inputs, and outputs
- [ ] Recorded questions for the client
- [ ] Created a flowchart with labeled decision paths
- [ ] Tested three example situations
- [ ] Prepared to explain the design

Completed Example | Hotel Stay Pricing

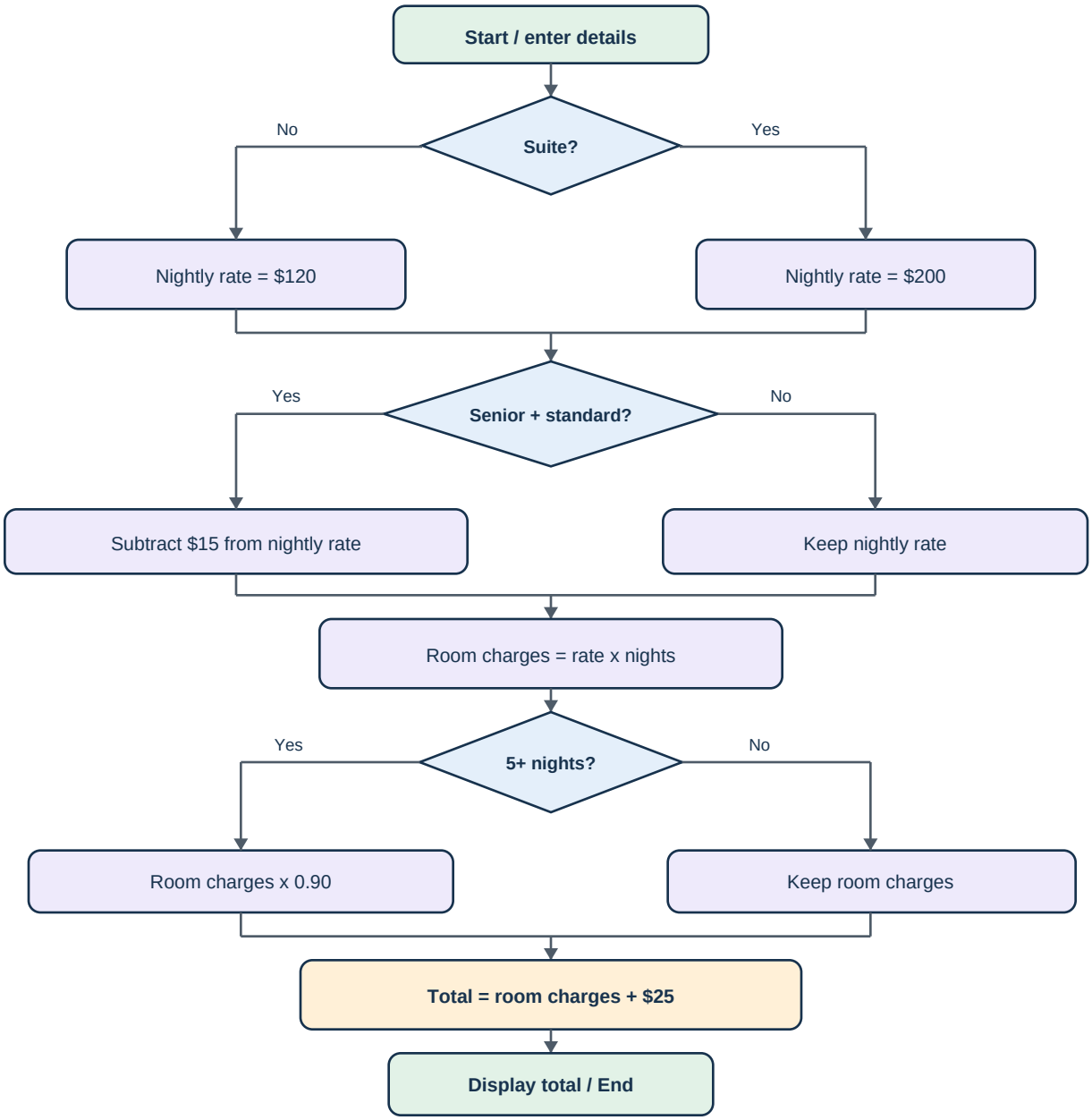
The hotel manager says:

"Standard rooms are $120 per night, and suites are $200. If someone stays five nights or more, we take 10% off their room charges. Seniors age 65 and older get $15 off each night in a standard room, but not a suite. Every reservation also has a one-time $25 cleaning fee."

Extracted rules

1. Standard: $120/night; suite: $200/night.
2. Seniors (65+) receive $15 off per night for standard rooms only.
3. Stays of 5+ nights receive 10% off room charges.
4. Add a one-time $25 cleaning fee after discounts.

Follow the flowchart: decisions choose which calculations to perform.



Worked example: Standard room, age 67, 6 nights:

($120 - $15) x 6 = $630; 10% off = $567; + $25 cleaning fee = $592.